

SQL

Presented By
Sylvia Anter



SQL



DQL

Select
Select' Family
1- from
2- where
3- Group by
4- having
5- Order by



DDL

1- create
2- alter
3- drop



DML

1- Insert
2- Update
3- Delete

Create , Alter , Drop

Create

Create Database
choose_name

create table
Choose_name



Alter

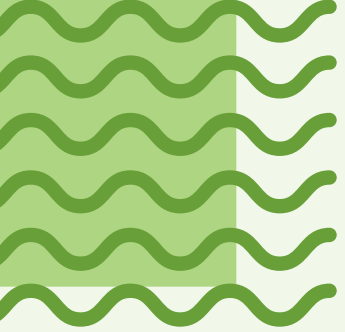
Alter table Ex_Sales
add column_name int



Drop

alter table Ex_Sales
Drop Column Column_name





Create



Create database Sales_Info

Create :



create table customers

```
(  
  CustomerKey int primary key ,  
  GeographyKey int ,  
  Customer varchar(30) Not null,  
  Continent varchar(10),  
  Region varchar(10)  
)
```

Create :



create table sales

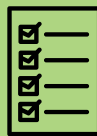
```
(  
  SON int primary key,  
  productkey int,  
  CustomerKey int,  
  qty float not null,  
  unitprice float not null,  
  revenue float not null,  
  OrderDate date,
```

```
  foreign key (productkey) references products(productkey),  
  foreign key (CustomerKey) references customers(CustomerKey)  
)
```

Insert , Update , Delete

Insert

Insert into
Table_name
values()



Update

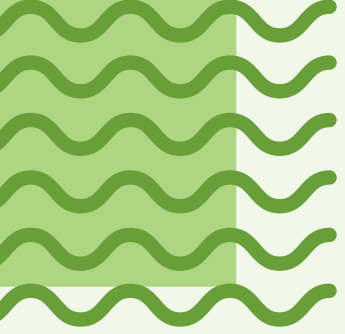
Update Table_name
Set column_name=new_value
where condition



Delete

Delete from Table_name
where Condition





Insert :



insert into customers
values
(111,110,'ahmed','africa','Egypt'),
(112,110,'ahmed','africa','soria'),
(113,110,'ahmed','africa','dubi')

Delete :



Delete from customers
where CustomerKey = 111

Update :



update customers
set Customer = 'olaa'
where CustomerKey = 112

customer_id	ref_id	name	region	country
111	110	ahmed	africa	Egypt
112	110	ahmed	africa	soria
113	110	ahmed	africa	soria
113	110	ahmed	africa	dubi

Select Family

Select

1- from

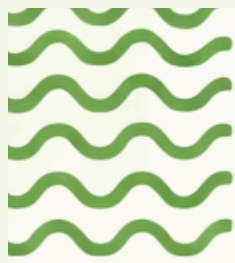
2- where

3- Group by

4- having

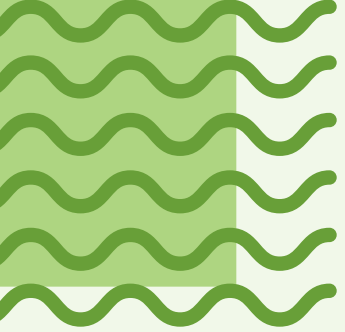
5- Order by





Sample Employee Table

id	first_name	last_name	salary	sex
10	Ahmed	Ali	4000	M
20	Sara	Mohamed	6000	F
30	Omar	Hassan	9000	M
40	Mona	Ibrahim	12000	F
50	Khaled	Youssef	7000	M
60	Nour	Adel	3000	F
70	Hany	Fathy	8500	M
Nulla	Laila	Samir	5000	F



Select :



Select *
from Employee

Select :



Select frist_name , Last_Name
from Employee

Select :



Select frist_name , Last_Name
from Employee
where salary > 5000

Select :



Select frist_name , Last_Name
from Employee
where first_name = "Ahmed"

Select :

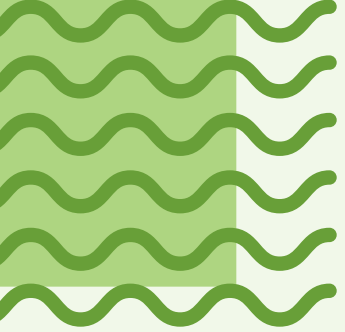


Select frist_name , Last_Name
from Employee
where salary between 5000 and 9000

Select :



Select frist_name , Last_Name
from Employee
where salary > 5000 and Sex = "M"



Select :



```
Select frist_name , Last_Name  
from Employee  
where salary not between 5000 and 9000
```

Select :



```
Select frist_name , Last_Name  
from Employee  
where salary > 5000 or Sex = "M"
```

Select :



```
Select frist_name , Last_Name  
from Employee  
where id in (10,20,30)
```

Select :



```
Select frist_name , Last_Name  
from Employee  
where id not in (10,20,30)
```

Select :

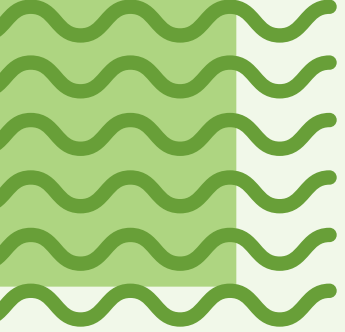


```
Select frist_name , Last_Name  
from Employee  
where id is Null
```

Select :



```
Select frist_name , Last_Name  
from Employee  
where id is not Null
```



Select :



```
Select frist_name , Last_Name  
from Employee  
where first_name like 'a%'
```

Select :



```
Select frist_name , Last_Name  
from Employee  
where first_name like '%d'
```

Select :



```
Select frist_name , Last_Name  
from Employee  
where first_name like '%a%'
```

Select :



```
Select frist_name , Last_Name  
from Employee  
where first_name like '_a%'
```

Select :

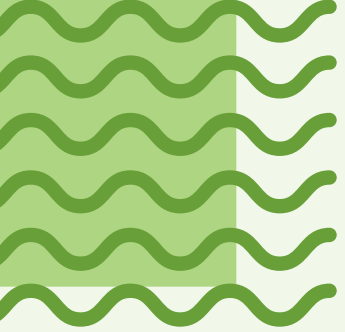


```
Select frist_name , Last_Name  
from Employee  
where first_name like '___a%'
```

Select :



```
Select frist_name , Last_Name  
from Employee  
Order by salary
```



Select :



```
Select frist_name , Last_Name  
from Employee  
where first_name like 'a%'
```

Select :



```
Select frist_name , Last_Name  
from Employee  
where first_name like '%d'
```

Select :



```
Select frist_name , Last_Name  
from Employee  
where first_name like '%a%'
```

Select :



```
Select frist_name , Last_Name  
from Employee  
where first_name like '_a%'
```

Select :

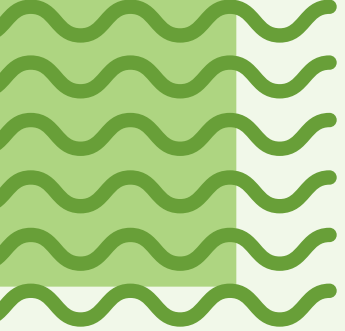


```
Select frist_name , Last_Name  
from Employee  
where first_name like '___a%'
```

Select :



```
Select frist_name , Last_Name  
from Employee  
Order by salary
```



Select :



Select min(Salary)
from employee

3000

Select :



Select max(Salary)
from employee

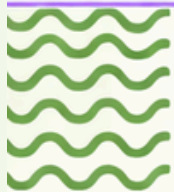
12000

Select :



Select sex , sum(Salary)
from employee
grouby by sex

SEX	Sum (Salary)
Male	10,000
Female	20,000



Sample Employee Table

id	first_name	last_name	salary	sex
10	Ahmed	Ali	4000	M
20	Sara	Mohamed	6000	F
30	Omar	Hassan	9000	M
40	Mona	Ibrahim	12000	F
50	Khaled	Youssef	7000	M
60	Nour	Adel	3000	F
70	Hany	Fathy	8500	M
Nulla	Laila	Samir	5000	F

Types Of Join



Inner Join



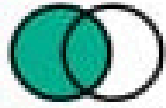
Left Join & Right Join



Full Join

Join Types

Left Outer Join



Key	T1
B	#
C	@
D	=
E	\$

Left Table

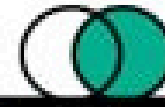
Key	T1	T2
B	#	15
C	@	20
D	=	
E	\$	

Key	T2
A	10
B	15
C	20
F	60

Right Table

All rows from Left + **matching** rows from Right

Right Outer Join



Key	T1
B	#
C	@
D	=
E	\$

Left Table

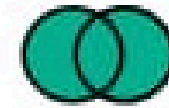
Key	T1	T2
A		10
B	#	15
C	@	20
F		60

Key	T2
A	10
B	15
C	20
F	60

Right Table

All rows from Right + **matching** rows from Left

Full Outer Join



Key	T1
B	#
C	@
D	=
E	\$

Left Table

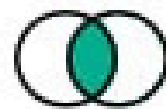
Key	T1	T2
A		10
B	#	15
C	@	20
D	=	
E	\$	
F		60

Key	T2
A	10
B	15
C	20
F	60

Right Table

All rows from both

Inner Join



Key	T1
B	#
C	@
D	=
E	\$

Left Table

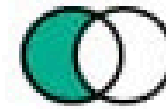
Key	T1	T2
B	#	15
C	@	20

Key	T2
A	10
B	15
C	20
F	60

Right Table

Matching rows between Left and Right

Left Anti Join



Key	T1
B	#
C	@
D	=
E	\$

Left Table

Key	T1	T2
D	=	
E	\$	

Key	T2
A	10
B	15
C	20
F	60

Right Table

Rows from Left **not matching** with Right

Right Anti Join



Key	T1
B	#
C	@
D	=
E	\$

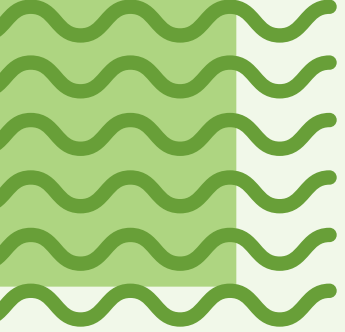
Left Table

Key	T1	T2
A		10
F		60

Key	T2
A	10
B	15
C	20
F	60

Right Table

Rows from Right **not matching** with Left



Inner Join :



```
select *  
from customers join sales  
on customers.CustomerKey = sales.customerkey
```

left join :



```
select *  
from customers left join sales  
on customers.CustomerKey = sales.customerkey
```

Right Join :

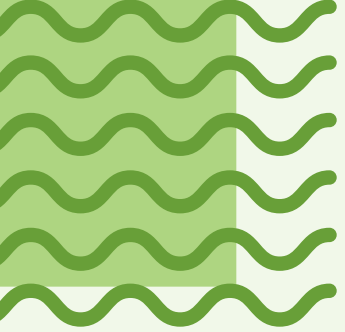


```
select *  
from customers right join sales  
on customers.CustomerKey = sales.customerkey
```

Full Join:



```
select *  
from customers full join sales  
on customers.CustomerKey = sales.customerkey
```

Select :



`select Size , sum(revenue) as "total_Revenue"`
`from products inner join sales`
`on products.ProductKey=sales.productkey`
`group by Size`
`having sum(revenue) > 15000`

Size	Sum (Revenue)
M	10,000
L	20,000

Select :



`select Continent , sum(revenue) as "total_Revenue"`
`from customers inner join sales`
`on customers.CustomerKey=sales.customerkey`
`group by Continent`
`having sum(revenue) > 150,000`

Continent	Sum (Revenue)
Africa	100,000
Eurpoe	250,000

SubQuery



```
SELECT column_name  
FROM table_name  
WHERE column_name = (  
    SELECT column_name  
    FROM another_table  
)
```

DeptID	Salary	Name	EmpID
1	8000	Ahmed	1
2	12000	Sara	2
1	10000	Mona	3

```

SELECT Name, Salary
FROM Employees
WHERE Salary >
(
SELECT AVG(Salary)
FROM Employees
)

```

عايزين نجيب الموظفين اللي مرتبتهم أعلى من
متوسط المرتبات

get the employees that their salaries
greater than the average of salaries

DeptID	Salary	Name	EmpID
1	8000	Ahmed	1
2	12000	Sara	2
1	10000	Mona	3

```
SELECT Name
FROM Employees
WHERE DeptID IN (
    SELECT DeptID
FROM Departments
WHERE ManagerID IS NOT NULL
)
```

عايزين الموظفين اللي شغالين في الأقسام اللي
فيها مديرين

get the employees that their department
has manager .

Project



create new data base
upload your file
analysis the data



show salesdetails table , salesReturn table

select *

from SalesDetails

select *

from SalesReturns



get only the customer that make orders

select Customer

from SalesDetails inner join Customers

on Customers.CustomerKey = SalesDetails.CustomerKey



get the customers that make returns

```
select Customer
from SalesReturns inner join Customers
on Customers.CustomerKey = SalesReturns.CustomerKey
```



get the customers that not make Orders

```
select Customer , SalesDetails.CustomerKey as "id_SD", Customers.CustomerKey
from Customers left join SalesDetails
on Customers.CustomerKey = SalesDetails.CustomerKey
where SalesDetails.CustomerKey is null
```



get the top (3) returned Product

```
select top (3) *  
from Product join SalesReturns  
on Product.ProductKey = SalesReturns.ProductKey  
order by ReturnQuantity desc
```



get the top (5) customers make return

```
select top (5) *  
from Customers join SalesReturns  
on Customers.CustomerKey = SalesReturns.CustomerKey  
order by ReturnQuantity desc
```




unique customer ?

```
select distinct customerkey  
from salesdetails
```



what is the total number of customer ?

```
select count(distinct customerkey)  
from salesdetails
```



what is the total number of customer ?

```
select count(distinct customerkey)  
from customers
```



what is the total revenue ?

```
select sum(totalamount)  
from salesdetails
```



what is the average of qty sold ?

```
select AVG(orderquantity)  
from salesdetails
```



get the biggest product according to QTY

```
select MAX(orderquantity)  
from salesdetails
```



get the smallest product according to QTY

```
select MIN(orderquantity)  
from salesdetails
```



what is the revenue Per continent

```
select continent , sum(totalamount)  
from customers join salesdetails  
on customers.customerkey = salesdetails.customerkey  
group by continent
```



what is the total qty sold by each category

```
select categoryname , sum(totalamount)  
from product join salesdetails  
on product.productkey = salesdetails.productkey  
group by categoryname
```



get the total revenue in each year

```
select year(orderdate) , sum(totalamount)
      from salesdetails
group by year(orderdate)
```



get the total qty in each month

```
select month(orderdate) , sum(totalamount)
      from salesdetails
group by month(orderdate)
order by month(orderdate)
```



get the orders amount/price that is greater than the average of total revenue / total amount

```
select salesdetailskey
from salesdetails
where totalamount >
(
select AVG(totalamount)
from salesdetails
)
```




get the product name , its sold qty , its returned qty

```
select productname , orderquantity , ReturnQuantity  
      from product join SalesDetails  
    on Product.ProductKey = SalesDetails.ProductKey  
      join SalesReturns  
    on Product.ProductKey = SalesReturns.ProductKey
```



use CTE (Common Table Expression)

```
with new_table as
(
select productname , orderquantity , ReturnQuantity
from product join SalesDetails
on Product.ProductKey = SalesDetails.ProductKey
join SalesReturns
on Product.ProductKey = SalesReturns.ProductKey
)
select ProductName , avg(orderquantity) OR_Q
from new_table
group by ProductName
```

DeptID	Salary	Name	EmpID
1	8000	Ahmed	1
2	12000	Sara	2
1	10000	Mona	3

```
SELECT Name, Salary,  
       CASE  
WHEN Salary >= 10000 THEN 'High Salary'  
       ELSE 'Low Salary'  
END AS SalaryCategory  
  
FROM Employees
```

عايزين نصنف المرتبات

SalaryCategory	Salary	Name
Low Salary	8000	Ahmed
High Salary	12000	Sara
High Salary	10000	Mona



--use case = if condition

with new_table as

(

select productname , orderquantity , ReturnQuantity

from product join SalesDetails

on Product.ProductKey = SalesDetails.ProductKey

join SalesReturns

on Product.ProductKey = SalesReturns.ProductKey

)

select ProductName , sum(orderquantity) O_Qty , sum(ReturnQuantity) R_Qty,

case

when sum(orderquantity) > sum(ReturnQuantity) then 'Safe'

when sum(orderquantity) < sum(ReturnQuantity) then 'not safe'

when sum(orderquantity) = sum(ReturnQuantity) then 'check'

end status_product

from new_table

group by ProductName